

Manuscript outline

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Introduction

1. The most frequently observed biological impact of climate change is major changes in phenology^{1,2,3,4,5}
 - (a) Shifts in spring and autumn phenology modifies when seasons starts and ends
 - (b) Understanding the consequences of these shifts requires understanding how much and why growing seasons have changed⁶.
 - (c) Spring phenology advances,^{7,8,9} driven by temperature^{10,2,11}
 - (d) Autumn phenology delays, but much less than spring^{12,13} and driven by temperature and photoperiod^{14,15,16,17}; shifts are weaker than springs^{12,13}
2. Shifts in spring and autumn phenology supports the assumption that longer growing seasons increase tree growth^{18,19}
 - (a) Recent research shows that this may not be the case^{20,21,22}
 - (b) Cabon showed at this decoupling at the global scale and Dow showed it locally.
 - (c) Dow showed earlier growth onset did not increase growth rate nor over all yearly growth increment.
 - (d) This has possible consequences on forest carbon cycle projections^{23,24}.
 - (e) The different effects that shifting spring and fall events have on growth could impact these projections
3. Understanding these finds requires answering why trees do not grow more despite longer growing seasons.
 - (a) Yes
 - (b) We don't understand how carbon is allocated to above-ground biomass, despite it being one of the largest carbon sink
 - (c) Trees may be source or sink limited, which emphasizes the importance of understanding the temperature sensitivity relationship between growth and photosynthesis
 - (d) Growth may be more sensitive to varying environmental conditions than photosynthesis, which could mislead models.
 - (e) Linear representation of growth as a function of carbon assimilation is wrong
 - (f) Hard nature of climate change makes understanding the internal physiological constraints and external constraints to growth critical
4. How trees internal programming and their susceptibility to environmental conditions could constrain their growth
 - (a) Growth as any other biological mechanism is genetically controlled.
 - (b) Tree's growth plasticity is strongly internally regulated, so trees can potentially just not increase growth by that much.

- (c) While CO₂ concentrations increased, other factors can be limiting.
 - (d) Droughts during longer seasons can offset benefits of an extended period to grow.
 - (e) Competition for nutrients, water and light can also offset a tree's capacity to take advantage of extra days in the season
5. Understanding the constraints to growth requires defining the growing season and growth
- (a) Different definitions of GS: we propose to use calendar (number of days) and thermal season (heat accumulation)
 - (b) The calendar season explicitly assumes that the number of days in the season is what matters in trees responses to growth, underlying that the growth rate remains relatively stable throughout the season. Accounting for just the days, may reveal the role of some neglected cooler days in the season that are of low importance for the thermal season.
 - (c) Xylogenesis throughout the growing season
 - (d) Tree rings as a proxy for growth
6. Growth drivers differences across species need to be considered
- (a) Phenology and the relationship between growth and growing season length varies heavily accross species
 - (b) Pooling all species together in carbon sequestrations models is a weakness
 - (c) We propose to use ground-based observations to address this issue
 - (d) These are useful to for the growth temporality of trees not suitable for experimental trials
 - (e) Cambial and leaf phenology are closely synchronized, so we can use them to as a proxy for start and end of season
7. Local adaptation could inform growth responses of species under future climate change.
- (a) Provenance refers to the geographic and climatic origin of a population.
 - (b) Trees from northern provenances sometimes have later spring phenology (Zeng2024Weak) and often earlier autumn phenology, thus a shorter growing season than southern provenance (*Vittasse2014TheInteraction and many other papers)
 - (c) This is especially true for budset, with the phenotypic variance in budset—which marks the end of primary growth—being strongly related to provenance with photoperiod acting as the primary cue (*Way2015Photoperiod)
 - (d) Thus, common gardens with provenances across a latitudinal gradient can inform tree's growth responses to future climatic conditions.
8. We will answer if trees grow more with longer growing seasons in a common garden using juvenile deciduous trees.
- (a) Determine if trees grow more with longer thermal vs calendar season (ref: conceptual figure)
 - (b) Understand the calendar growing season relationship with growth and how it relates to the SOS and EOS
 - (c) Determine how growth changes across species and their provenances
 - (d) We will look at this using four species from four provenances across three years using both leaf phenology and tree rings.

Methods

Results

Separate document

Discussion

1. What we did, how we did it and what we found
 - (a) We used 81 juvenile trees of four species across four provenances to understand if longer growing seasons using leaf phenology increased growth using tree rings
 - (b) We found that longer growing seasons increased growth for all species, but varied depending on the metric
 - (c) Thermal season was a better predictor for growth than the calendar season
2. Thermal season being a better predictor than the calendar season may be caused by it weighting the relative importance of the days on growth better than the calendar season
 - (a) Growth rate follows a temperature response curve, thus trees don't grow at the same speed all year round
 - (b) Cool and short spring days count for less in the thermal season compared to the calendar season where each day counts the same
3. Disentangling the calendar season with the start and end of season may reveal something...
 - (a) 123
4. Our trees grew more with longer seasons in absence of external constraints
 - (a) Droughts can strongly impact tree growth, thus when seasons are longer, but not drier, trees just may be able to fully take advantage of these extra days
 - (b) Earlier spring means trees start water uptake earlier, which may increase the severity of droughts later in the season
 - (c) Dow showed that growth and growing season length are decoupled in a natural system where trees compete for water, but also for light and nutrients
 - (d) Though we did not control for nutrients, our study design allowed us to control for light availability where taller trees did not shade the others at the life stage we looked at
 - (e) The timing of phenological phases are shifting and this will affect inter-specific competition for resources, this highlights the importance of accounting for competition when adresssing this question
5. Our species have different life history strategies and this may inform how they respond to growing season length
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6. We found little effect of provenance, suggesting local adaption on growth may be weak under current conditions
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7. Our study has limitations
 - (a) We had low sample size and not individuals for each provenance for all species
 - (b) We didn't account for extreme temperatures in our models, though there was no extended heat wave periods throughout the study
8. Our results highlights the importance of accounting for
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Supplemental results